

Separable & Linear Equations: Sorting the Mess, Then Zipping It Shut

A ground-up companion to Lecture 3 — how to split a separable equation into two independent pieces, and how the integrating factor turns a linear equation's messy left side into one clean derivative you can integrate directly.

0 The Big Idea: Two Different Tricks, Two Different Shapes

TL;DR

This lecture hands you two of the most useful "recognize the shape, apply the trick" methods for first-order ODEs. **Separable** equations let you physically slide all the y -stuff to one side and all the x -stuff to the other, then integrate each side on its own. **Linear** equations don't split apart like that — instead, you multiply the whole equation by a special helper function that collapses the left side into a single clean derivative.

Both methods solve first-order equations $dy/dx = (\text{something})$, but they only work when the right-hand side has a specific shape. Spotting which shape you're looking at is 90% of the battle:

- **Separable:** the right side factors into $g(x) \cdot h(y)$ — a pure- x piece times a pure- y piece.
- **Linear:** the equation can be written as $dy/dx + P(x)y = f(x)$ — y and its derivative appear only to the first power, never multiplied together or squared.

Neither method cares about the other's equations — a linear equation is usually *not* separable, and a separable equation is usually *not* linear. They're two different keys for two different locks.

1 Separable Equations: Splitting the Mixture

TL;DR

If $dy/dx = g(x)h(y)$, you can legally slide every y (and dy) to one side and every x (and dx) to the other. Once they're apart, integrate each side completely independently.

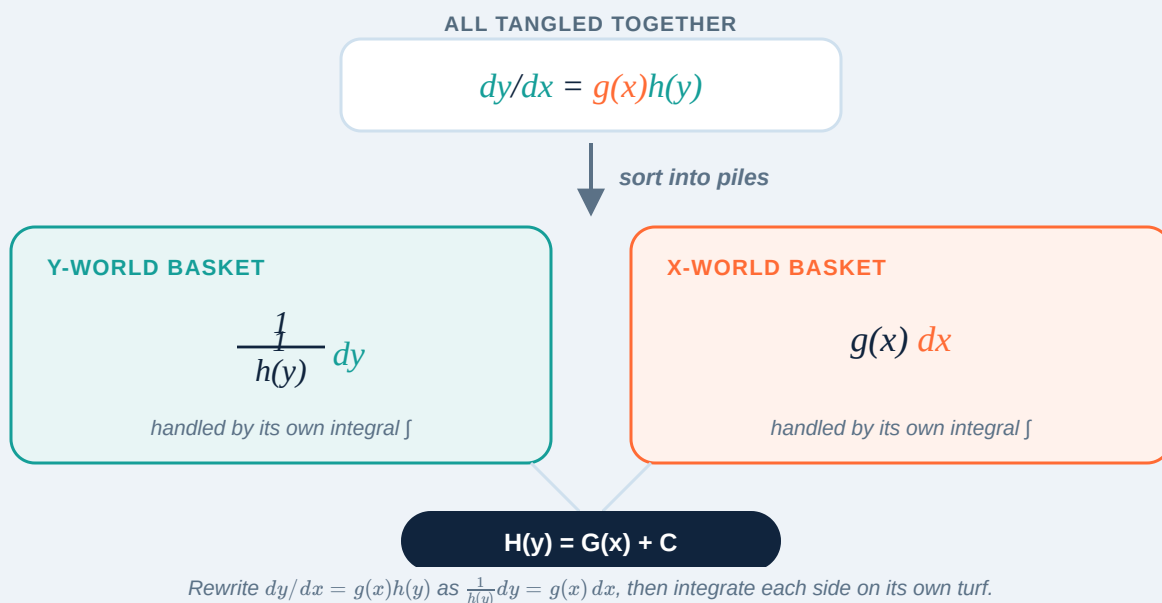
A first-order DE is called **separable** if it can be written as:

$$\frac{dy}{dx} = g(x) h(y)$$

For example, $dy/dx = xy^2$ is separable ($g(x) = x$, $h(y) = y^2$), and so is $dy/dx = \sin x e^y$. An equation like $dy/dx = x + y$ is *not* separable — there's no way to factor $x + y$ into a pure- x part times a pure- y part.

THE ANALOGY

Imagine the equation as a basket of laundry where x -colored shirts and y -colored socks got mixed together in the wash. You can't clean (integrate) a mixed pile all at once — you have to **sort first**: every sock goes in the y -basket, every shirt goes in the x -basket. Once sorted, each basket runs through its own wash cycle (its own integral) completely independently of the other. At the end, you don't literally combine the piles back together — you just note that both baskets finished on the same load, which is what the shared constant C remembers.



Formally, the method of solution is just algebra plus two independent integrals:

STEP 1 Divide both sides by $h(y)$ (as long as $h(y) \neq 0$) so every y is on the left:

$$\frac{1}{h(y)} \frac{dy}{dx} = g(x).$$

STEP 2 Treat dy/dx like a fraction and multiply both sides by dx : $\frac{1}{h(y)} dy = g(x) dx$.

Everything with a y (including dy) is now on the left; everything with an x (including dx) is on the right.

STEP 3 Integrate both sides — the left with respect to y , the right with respect to x :

$$\int \frac{1}{h(y)} dy = \int g(x) dx \implies H(y) = G(x) + C.$$

ONE CATCH WORTH REMEMBERING

Dividing by $h(y)$ silently assumes $h(y) \neq 0$. Any constant value $y = y_0$ where $h(y_0) = 0$ is also a perfectly valid solution (it makes $dy/dx = 0$ on both sides) — it just gets lost in the division step and has to be checked separately. You'll see exactly this happen in Example 3 below.

2 Worked Examples: Separable Equations

EXAMPLE 1

Solve $(1+x)dy - ydx = 0$.

SEPARATE Rearrange to $(1+x)dy = ydx$, then divide by $y(1+x)$: $\frac{dy}{y} = \frac{dx}{1+x}$.

INTEGRATE $\int \frac{dy}{y} = \int \frac{dx}{1+x} \implies \ln|y| = \ln|1+x| + C_1$

SOLVE FOR y Exponentiate both sides and absorb e^{C_1} into a fresh constant C : $y = C(1+x)$.

SOLUTION

$$y = C(1+x)$$

EXAMPLE 2 — AN IVP

Solve $\frac{dy}{dx} = -\frac{x}{y}$, subject to $y(4) = -3$.

SEPARATE Cross-multiply: $y dy = -x dx$.

INTEGRATE $\int y dy = -\int x dx \implies \frac{y^2}{2} = -\frac{x^2}{2} + C_1 \implies x^2 + y^2 = C$

APPLY THE IC Plug in $x = 4, y = -3$: $16 + 9 = C \implies C = 25$.

PARTICULAR SOLUTION

$$x^2 + y^2 = 25$$

NOTICE

The general solution $x^2 + y^2 = C$ is a family of circles centered at the origin — the IVP just picks out the one radius-5 circle passing through $(4, -3)$.

3 Worked Example 3 — and a Callback to Lecture 2

EXAMPLE 3

Solve $\frac{dy}{dx} = y^2 - 4$.

The right side is $g(x)h(y)$ with $g(x) = 1$ and $h(y) = y^2 - 4$ — separable. But watch for the catch from the previous page: $h(y) = 0$ at $y = \pm 2$, so those two constant functions are solutions too, and they'll need to be added back in at the end.

SEPARATE $\frac{dy}{y^2 - 4} = dx$

PARTIAL FRACTIONS $\frac{1}{y^2 - 4} = \frac{1}{(y - 2)(y + 2)} = \frac{1}{4} \left[\frac{1}{y - 2} - \frac{1}{y + 2} \right]$

INTEGRATE $\frac{1}{4} \ln \left| \frac{y - 2}{y + 2} \right| = x + C_1 \implies \ln \left| \frac{y - 2}{y + 2} \right| = 4x + C_2$

SOLVE FOR y Exponentiate: $\frac{y - 2}{y + 2} = Ce^{4x}$, then clear the fraction and isolate y :

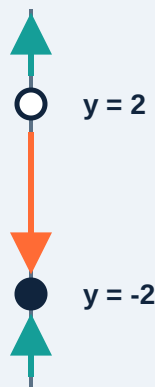
$$y = \frac{2(1 + Ce^{4x})}{1 - Ce^{4x}}$$

SOLUTION

$$y = \frac{2(1 + Ce^{4x})}{1 - Ce^{4x}} \quad \text{together with the two "lost" constant solutions } y = 2 \text{ and } y = -2$$

WHY THIS CONNECTS BACK TO LECTURE 2

$dy/dx = y^2 - 4$ doesn't have x on the right at all — it's an **autonomous** equation, exactly the kind Lecture 2 analyzed with phase lines. The two "lost" constant solutions $y = 2$ and $y = -2$ are precisely its **critical points**. The phase-line picture below tells you, at a glance, which way nearby solutions drift — without solving a single integral.



$y = 2$ — UNSTABLE

Nearby paths drift away on both sides.

$y = -2$ — STABLE

Nearby paths are pulled toward it.

Sign of $y^2 - 4$ in each region tells you the arrow direction: positive means y increases (arrow up), negative means y decreases (arrow down).

READING THE ARROWS

For $y > 2$: $y^2 - 4 > 0$, so y increases — solutions run away upward. For $-2 < y < 2$: $y^2 - 4 < 0$, so y decreases — solutions drift down toward $y = -2$. For $y < -2$: $y^2 - 4 > 0$ again, so y increases — solutions climb back up toward $y = -2$. Every solution in the middle band or below is pulled toward -2 ; that's why it's **stable**, while $y = 2$ pushes everything away and is **unstable**.

4 Linear Equations: The Zipper Trick

TL;DR

A linear equation can be forced into standard form $dy/dx + P(x)y = f(x)$. Multiplying the whole thing by the **integrating factor** $\mu(x) = e^{\int P(x) dx}$ makes the left side collapse — by the product rule, in reverse — into the single clean derivative $\frac{d}{dx}[\mu(x)y]$, which you can just integrate.

A first-order DE is **linear** if it has the form:

$$a_1(x) \frac{dy}{dx} + a_0(x) y = g(x)$$

Dividing through by $a_1(x)$ puts it in the more useful **standard form**:

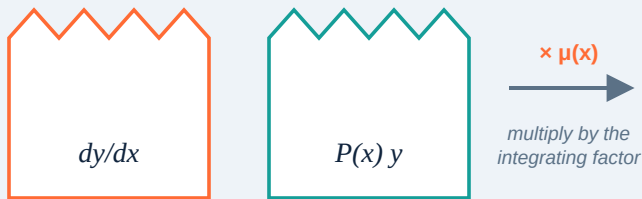
$$\frac{dy}{dx} + P(x) y = f(x)$$

Unlike separable equations, you generally can't slide the y 's away from the dy/dx here — $P(x)y$ is stuck next to dy/dx with no way to factor them apart. This method takes a completely different route: instead of splitting the equation, it *reshapes* it.

THE ANALOGY

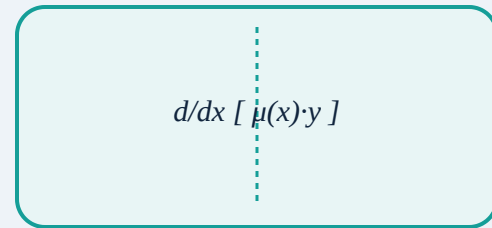
Think of dy/dx and $P(x)y$ as the two loose rows of teeth on an open zipper — close, but not connected, and you can't treat them as one thing. The **integrating factor** $\mu(x)$ is the zipper pull: multiply every term by it, and the two loose pieces suddenly interlock into one smooth, continuous seam — $\frac{d}{dx}[\mu(x)y]$. That seam is something you already know how to handle: just integrate it.

BEFORE — TWO LOOSE TEETH



Two separate terms — no clean single derivative.

AFTER — ONE SMOOTH SEAM



One clean product-rule derivative — ready to integrate.

The product rule, run backward: $\frac{d}{dx}[\mu y] = \mu \frac{dy}{dx} + \mu' y$. Choosing $\mu' = \mu P(x)$ is exactly what makes the two loose terms zip together.

WHERE $M(X) = E^{\int P(X)DX}$ ACTUALLY COMES FROM

We want a function $\mu(x)$ so that $\mu \frac{dy}{dx} + \mu P(x)y$ is exactly $\frac{d}{dx}[\mu y] = \mu \frac{dy}{dx} + \mu' y$. Matching the second terms requires $\mu' = \mu P(x)$, a tiny separable equation in μ itself! Solving it ($d\mu/\mu = P(x)dx$) gives $\ln \mu = \int P(x)dx$, i.e. $\mu(x) = e^{\int P(x)dx}$.

Putting it together, the general solution is $y = y_c + y_p$, where:

$$y_c = c e^{-\int P(x)dx} \quad y_p = e^{-\int P(x)dx} \int e^{\int P(x)dx} f(x) dx$$

In practice, nobody memorizes that formula — you rebuild it every time with three mechanical steps:

STEP I Put the equation in standard form $dy/dx + P(x)y = f(x)$, read off $P(x)$, and build the integrating factor $\mu(x) = e^{\int P(x)dx}$.

STEP II Multiply every term of the standard-form equation by $\mu(x)$. The left side is now guaranteed to collapse into $\frac{d}{dx}[\mu(x)y]$.

STEP III Integrate both sides with respect to x , then divide by $\mu(x)$ to solve for y .

EXAMPLE 4

Solve $\frac{dy}{dx} - 3y = 6$.

STEP I Already in standard form: $P(x) = -3$, $f(x) = 6$. Integrating factor: $\mu(x) = e^{\int -3 dx} = e^{-3x}$.

STEP II Multiply through by e^{-3x} : $e^{-3x} \frac{dy}{dx} - 3e^{-3x}y = 6e^{-3x} \implies \frac{d}{dx} [e^{-3x}y] = 6e^{-3x}$

STEP III Integrate both sides: $e^{-3x}y = -2e^{-3x} + C$. Divide by e^{-3x} : $y = -2 + Ce^{3x}$.

SOLUTION

$$y = -2 + Ce^{3x}$$

EXAMPLE 5

Solve $(x^2 - 9)\frac{dy}{dx} + xy = 0$.

STEP I Divide by $(x^2 - 9)$ to reach standard form: $\frac{dy}{dx} + \frac{x}{x^2 - 9}y = 0$, so $P(x) = \frac{x}{x^2 - 9}$.

BUILD μ $\int \frac{x}{x^2 - 9} dx = \frac{1}{2} \ln |x^2 - 9|$, so $\mu(x) = e^{\frac{1}{2} \ln |x^2 - 9|} = \sqrt{x^2 - 9}$.

STEP II Multiply through by $\sqrt{x^2 - 9}$: the left side collapses to $\frac{d}{dx} [\sqrt{x^2 - 9}y] = 0$.

STEP III Integrate: $\sqrt{x^2 - 9}y = C$, so $y = \frac{C}{\sqrt{x^2 - 9}}$.

SOLUTION

$$y = \frac{C}{\sqrt{x^2 - 9}}$$

SIDE NOTE

This particular equation is secretly separable too ($dy/y = -x/(x^2 - 9) dx$) — a nice reminder that the two categories from Section 0 can overlap. When they do, either method gets you to the same answer.

6

Bonus: Piecewise-Linear Equations — a Relay Race

TL;DR

When the "input" $f(x)$ switches rules partway through (a piecewise function), solve the linear equation separately on each piece, then hand off the baton: force the ending value of one piece to match the starting value of the next.

Some real models (especially in biology and engineering) have a coefficient or forcing term that's only piecewise-defined — a switch that's on, then off. The fix is simple in spirit: solve the IVP one interval at a time, and glue the pieces together using **continuity** at the seam.

THE ANALOGY

Picture a relay race with two runners. Runner 1 (the equation on $0 \leq x \leq 1$) starts at the official starting line ($y(0) = 0$) and runs until the handoff zone at $x = 1$. Runner 2 (the equation on $x > 1$) doesn't get to start from zero — they have to start exactly where Runner 1 left off. That handoff condition, "Runner 2's starting value = Runner 1's final value," is the only thing that links the two pieces together.

EXAMPLE 6

Solve $\frac{dy}{dx} + y = f(x)$, $y(0) = 0$, where $f(x) = \begin{cases} 1, & 0 \leq x \leq 1 \\ 0, & x > 1 \end{cases}$

PIECE 1 ($0 \leq x \leq 1$) $dy/dx + y = 1$. Integrating factor $\mu = e^x$: $\frac{d}{dx}[e^x y] = e^x \implies e^x y = e^x + C_1 \implies y = 1 + C_1 e^{-x}$. Apply $y(0) = 0$: $C_1 = -1$.

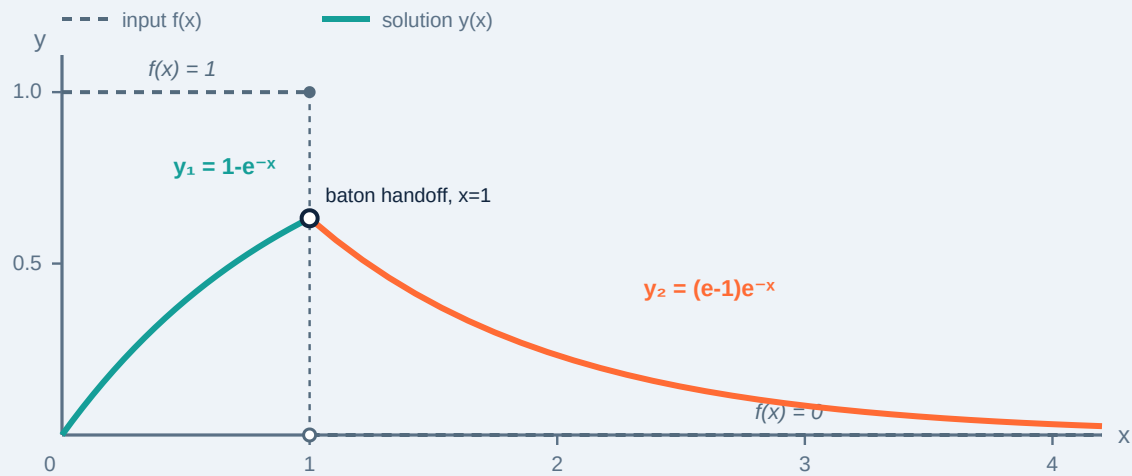
RESULT $y_1(x) = 1 - e^{-x}$, valid for $0 \leq x \leq 1$. At the handoff point, $y_1(1) = 1 - e^{-1} \approx 0.632$.

PIECE 2 ($x > 1$) $dy/dx + y = 0$. Same $\mu = e^x$: $\frac{d}{dx}[e^x y] = 0 \implies y = C_2 e^{-x}$.

HAND OFF THE BATON Force $y_2(1) = y_1(1)$: $C_2 e^{-1} = 1 - e^{-1} \implies C_2 = e - 1$.

SOLUTION

$$y(x) = \begin{cases} 1 - e^{-x}, & 0 \leq x \leq 1 \\ (e - 1)e^{-x}, & x > 1 \end{cases}$$



The dashed line is the on/off switch $f(x)$; the solid curve is the resulting $y(x)$ — smooth and continuous even though its formula changes at $x = 1$.

NOTICE

Even though $f(x)$ jumps abruptly at $x = 1$, the solution $y(x)$ doesn't jump — it's continuous. The handoff condition is exactly what guarantees that smoothness; only the *slope* of $y(x)$ has a kink there, not $y(x)$ itself.



Tying It All Together

Separable equations ($dy/dx = g(x)h(y)$) are solved by physically sorting the mixture: slide every y to one side, every x to the other, and integrate each pile independently. Watch for constant solutions lost in the division step — when the equation is also autonomous, those lost solutions are exactly the critical points from Lecture 2's phase-line analysis.

Linear equations ($dy/dx + P(x)y = f(x)$) can't be split apart, so instead you reshape them: multiply by the integrating factor $\mu(x) = e^{\int P(x)dx}$, and the left side snaps together into one clean derivative, $\frac{d}{dx}[\mu y]$, ready to integrate directly.

When the forcing term $f(x)$ itself is piecewise, there's no new theory needed — just solve the linear equation on each piece separately, then match values at the seam so the overall solution stays continuous.

A plain-English companion to: ENGR 213, Lecture 3 (Sept 16, 2026), Prof. A. Haghighat M., Concordia University — reference text: Zill, Advanced Engineering Mathematics (7th ed.), Sections 2.2–2.3. Worked solutions to Examples 1–6 and the analogies (laundry-sorting, phase-line callback, zipper integrating factor, relay-race handoff) are original, built to fill in and illustrate the same lecture material.